

The Schläfli Decomposition of the 600-cell: Five 24-cell Cosets and the Induced A_5 Action

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Reproducibility. The formal claims of this note — construction of V_{600} as the binary icosahedral group $2I$, identification of the σ -fixed subset as the binary tetrahedral group $2T$, the partition of V_{600} into five right cosets of $2T$, verification that each coset carries the intrinsic 24-cell distance structure, and the induced action of $2I$ on the five cosets descending through $\{\pm 1\}$ to a transitive A_5 on five points — are certified by exact $\mathbb{Q}(\sqrt{5})$ arithmetic in the accompanying repository. No floating-point computation enters any formal certificate; if any floating computations appear they are cross-checks only. One-command reproduction is described in Section 8.

Abstract

The 600-cell vertex set is represented exactly as the 120 unit icosian quaternions, i.e. the binary icosahedral group $2I$. The 24-cell vertex set is represented as the 24-element binary tetrahedral group $2T$, characterised as the σ -fixed subset of $2I$ under the Galois twist $\sigma: \sqrt{5} \mapsto -\sqrt{5}$. We prove that $2T$ has index 5 in $2I$, that its five right cosets partition V_{600} into five disjoint 24-element subsets, that each subset carries the intrinsic 24-cell distance structure (an 8-regular graph at squared distance 1, 96 edges), and that right multiplication by $2I$ induces a transitive action on the five cosets with kernel exactly $\{\pm 1\}$, so the image is $A_5 = 2I/\{\pm 1\}$ acting on five points. Each claim is certified by exact $\mathbb{Q}(\sqrt{5})$ arithmetic.

1 Introduction

The 600-cell is the regular 4-polytope $\{3, 3, 5\}$ with 120 vertices. The 24-cell is the regular 4-polytope $\{3, 4, 3\}$ with 24 vertices. Classically [Coxeter 1973, Conway–Sloane 1999], the vertex set of the 600-cell can be identified with the binary icosahedral group $2I$, the vertex set of the 24-cell with the binary tetrahedral group $2T$, and $2T$ sits inside $2I$ as a subgroup of index 5. Its cosets therefore give five copies of the 24-cell whose union is V_{600} — a finite-geometric decomposition that we make explicit, exact, and reproducible here.

This note supplies the finite-geometric substrate used by the 24–600 spectral bridge of [Smart 2026]. That spectral note asks a further question: whether the local Laplacian eigenspaces on the 24-cell sectors zero-extend into corresponding eigenspaces of the global 600-cell Laplacian. Here we prove and certify the underlying coset decomposition independently of any spectral claim.

Because $2T$ is not normal in $2I$, left and right cosets give different partitions of V_{600} ; we fix the right-coset convention $2T \cdot g$ throughout, matching the spectral bridge.

No physical claim is made. Within the broader closure-geometry programme that motivated this work, the decomposition is one of the finite geometric structures used downstream; in the present note it is a self-contained finite-group / finite-geometry result.

2 Exact construction of V_{600} as $2I$

We work in $\mathbb{Q}(\sqrt{5})$, represented as ordered pairs $(a, b) \mapsto a + b\sqrt{5}$ with $a, b \in \mathbb{Q}$. Set $\varphi = (1 + \sqrt{5})/2$. Define $V_{600} \subset \mathbb{Q}(\sqrt{5})^4$ as the 120-element set:

- 8 vertices $\pm e_i$, $i = 0, \dots, 3$;
- 16 vertices $(\pm 1, \pm 1, \pm 1, \pm 1)/2$;
- 96 vertices: even permutations of $(0, \pm 1/(2\varphi), \pm 1/2, \pm \varphi/2)$.

This vertex set is the standard *icosian model* of the 600-cell vertex set; under quaternion multiplication it is the binary icosahedral group $2I$ in its classical presentation [Coxeter 1973, Conway–Sloane 1999]. Theorem 1 verifies, by exact arithmetic, that the explicit 120-element model above does satisfy the group axioms (closure, identity, inverses) on this set; the identification with the abstract $2I$ is the classical reference, not a consequence of the certificate alone.

Theorem 1 (Exact group structure of the icosian model). *The 120 vectors above are pairwise distinct; each has squared $\mathbb{Q}(\sqrt{5})$ -norm exactly 1; the set contains the identity quaternion $(1, 0, 0, 0)$; it is closed under quaternion multiplication; and the conjugate of every element (which equals its inverse since norms are 1) is again in V_{600} . Hence V_{600} , equipped with quaternion multiplication, is a finite group of order 120. By [Coxeter 1973, Conway–Sloane 1999] this is the standard icosian model of $2I$.*

Proof by exact rational certificate. The accompanying repository certifies each clause by exact $\mathbb{Q}(\sqrt{5})$ arithmetic in the routine `certify_v600_construction` in module `closure_transform_engine.decomposition`. Distinctness is checked on $\mathbb{Q}(\sqrt{5})$ keys; unit norm componentwise; closure runs all $120^2 = 14\,400$ products $u \cdot v$ and verifies each lies in the key set; inverses run all 120 conjugates and verify membership. No floating-point arithmetic is used. $\square$

3 The σ -fixed 24-cell as $2T$

Let $\sigma: \mathbb{Q}(\sqrt{5}) \rightarrow \mathbb{Q}(\sqrt{5})$ be the Galois twist $\sqrt{5} \mapsto -\sqrt{5}$. We extend σ componentwise to $\mathbb{Q}(\sqrt{5})^4$. Define

$$V_{24} := \{v \in V_{600} : \sigma(v) = v\}.$$

The σ -fixed subset consists exactly of the 8 unit vectors $\pm e_i$ and the 16 half-integer vectors $(\pm 1, \pm 1, \pm 1, \pm 1)/2$ from Section 2, since these are precisely the vectors with no $\sqrt{5}$ component.

Theorem 2 (Exact subgroup structure of the σ -fixed set). $|V_{24}| = 24$. *The set $V_{24} \subset V_{600}$ is closed under quaternion multiplication, contains the identity, and contains the inverse of each of its elements. Hence V_{24} is a subgroup of V_{600} of order 24, and the index of V_{24} in V_{600} is $|V_{600}|/|V_{24}| = 120/24 = 5$. The 24-element set — the eight units $\pm e_i$ together with the sixteen half-integer vectors $(\pm 1, \pm 1, \pm 1, \pm 1)/2$ — is the standard Hurwitz-unit model of the binary tetrahedral group $2T$ [Coxeter 1973, Conway–Sloane 1999]; the certificate verifies that this explicit model satisfies the subgroup axioms inside V_{600} .*

Proof by exact rational certificate. `certify_v24_subgroup` computes the fixed set V_{24} in exact $\mathbb{Q}(\sqrt{5})$ arithmetic, verifies its size is 24, runs all $24^2 = 576$ products $u \cdot v$ for $u, v \in V_{24}$ and checks each lies in V_{24} , verifies that the identity is fixed, and verifies that the conjugate of every fixed element is fixed. The index follows from $|2I|/|2T|$. $\square$

4 The five right cosets

Convention. We work with the right cosets $2T \cdot g$ of $2T$ in $2I$. Concretely, for any $v \in V_{600}$ the orbit $2T \cdot v = \{h \cdot v : h \in 2T\}$ is the right coset containing v . Note that $2T$ is not normal in $2I$: left and right cosets give different partitions of $2I$, related by the inversion map $g \mapsto g^{-1}$ which exchanges the two coset families and fixes the identity coset $2T$ itself. We use right cosets to match the convention of the 24–600 spectral bridge artifact [Smart 2026].

Cosets. Label them C_0, C_1, C_2, C_3, C_4 with $C_0 = V_{24}$ itself. The remaining four cosets $C_1, \dots, C_4$ are the four V_{24} -orbits on the 96 σ -mobile vertices.

Theorem 3 (Schläfli decomposition). *The five right cosets of $2T$ in $2I$ partition V_{600} :*

$$V_{600} = C_0 \sqcup C_1 \sqcup C_2 \sqcup C_3 \sqcup C_4, \quad |C_i| = 24 \text{ for each } i,$$

$C_0 = 2T$, and $C_1 \cup C_2 \cup C_3 \cup C_4$ is exactly the set of 96 σ -mobile vertices. In particular, every vertex of V_{600} belongs to exactly one coset.

Proof by exact rational certificate. `right_cosets` and `certify_right_cosets` compute the five orbits, verify that there are exactly 5 of them with sizes 24 each, that they are pairwise disjoint, that their union is V_{600} , that $C_0 = V_{24}$, and that the union $C_1 \cup C_2 \cup C_3 \cup C_4$ equals the σ -mobile complement of V_{24} . Each membership check is an exact set operation on vertex indices. $\square$

5 Each coset carries the 24-cell structure

For any pair $u, v \in V_{600}$ define the squared $\mathbb{Q}(\sqrt{5})$ -distance $d^2(u, v) = |u - v|^2 \in \mathbb{Q}(\sqrt{5})$.

There are 8 distinct non-zero values of d^2 on V_{600} . The shortest is

$$d_{V_{600}}^2 = \frac{3 - \sqrt{5}}{2} \approx 0.382,$$

which defines the 600-cell edge graph (12-regular, 720 edges). Inside each coset C_i the shortest non-zero d^2 between members is the next shell up:

$$d_{24}^2 = 1.$$

Theorem 4 (Each coset is a 24-cell). *For each $i \in \{0, 1, 2, 3, 4\}$, $C_i = 2T \cdot g_i$ is isometric to $2T$ as a metric subspace of S^3 . In particular, the shortest non-zero squared distance between members of C_i is exactly 1, and the graph on C_i whose edges are the pairs at $d^2 = 1$ is the standard 24-cell edge graph (24 vertices, 96 edges, 8-regular). Furthermore, no pair of vertices within a single coset is joined by a V_{600} -edge: every pair within C_i at separation $d_{V_{600}}^2$ counts to zero.*

Proof. For any unit quaternion $g \in 2I$, right multiplication $v \mapsto v g$ is a Euclidean isometry of $\mathbb{R}^4$ restricted to S^3 : indeed $|v_1 g - v_2 g| = |(v_1 - v_2) g| = |v_1 - v_2|$ since $|g| = 1$. Hence the map $2T \rightarrow 2T \cdot g_i$, $h \mapsto h g_i$, is an isometric bijection; the intrinsic metric on $C_i = 2T \cdot g_i$ is the metric on $2T$ pulled across. Since the standard Hurwitz-unit model of $2T$ has shortest non-zero squared distance 1 and its $d^2=1$ graph is the 24-cell edge graph ([Coxeter 1973, Conway–Sloane 1999]), each coset C_i inherits exactly that structure.

The accompanying repository (`certify_each_coset_is_24cell`) additionally verifies, as a numerical check on the explicit coordinates, that for each coset: the minimum non-zero squared distance is exactly (1, 0) (i.e. 1); the count of directed $d^2=1$ pairs is 192 (so 96 undirected edges); every vertex has exactly 8 such neighbours inside the coset; and the count of intra-coset pairs at the V_{600} -shell distance $(3 - \sqrt{5})/2$ is zero. $\square$

Remark 5. Theorem 4 says that each coset is an isometric copy of the 24-cell, but the 24-cell edges live at the next V_{600} -distance shell, not at the shortest one. Equivalently, the five 24-cells are *not* induced subgraphs of the 600-cell edge graph: every 600-cell edge runs between two different cosets, and every intra-coset 24-cell edge runs at a strictly longer distance than a 600-cell edge. This separation is the structural feature that the 24–600 spectral bridge then exploits to compare local and global Laplacians.

6 Induced action on the five cosets

Right multiplication by $g \in 2I$ permutes the right cosets:

$$(2T \cdot v) \cdot g = 2T \cdot (vg).$$

This well-defined map from $2I$ to the symmetric group on the five cosets is the focus of this section.

Theorem 6 (A_5 coset action). *The induced map $2I \rightarrow \text{Sym}\{C_0, C_1, C_2, C_3, C_4\}$:*

1. *is transitive on the five cosets;*
2. *has image of size 60 (i.e. exactly 60 distinct coset-level permutations arise from the 120 elements of $2I$);*
3. *has kernel exactly $\{+1, -1\}$, the centre of $2I$;*
4. *lands entirely in the alternating group: every element of the image is an even permutation;*
5. *has image equal to $A_5 \subset S_5$, with conjugacy-class sizes $(1, 15, 20, 12, 12)$, matching the standard A_5 character table.*

Equivalently, the coset action factors through the surjection $2I \rightarrow 2I/\{\pm 1\} = I \cong A_5$.

Proof by exact rational certificate. `certify_a5_coset_action` enumerates all 120 vertex permutations P_h for $h \in 2I$, reads off the induced coset permutation $\pi(h) \in S_5$ (well-defined by Theorem 3), and computes:

- the number of distinct images $|\{\pi(h) : h \in 2I\}|$, which equals 60 exactly;
- the kernel $\{h : \pi(h) = \text{id}\}$, which contains exactly two elements — the identity quaternion and its negative, by direct exact equality of $\mathbb{Q}(\sqrt{5})$ entries;
- the parity of every element of the image, which is even in all 60 cases;
- the conjugacy-class partition of the image under its own conjugation, which has class sizes $(1, 15, 20, 12, 12)$ matching A_5 .

Transitivity follows from the orbit of C_0 under the image, which covers all five cosets. All 60 image permutations are verified even, so the image lies inside $A_5 \subset S_5$; since $|A_5| = 60$ and the image has 60 elements, the image equals all of A_5 . $\square$

7 Relationship to the 24–600 spectral bridge

This note proves the finite-geometric decomposition used by the 24–600 spectral bridge artifact [Smart 2026]. The spectral bridge then asks a further question: whether local eigenspaces of the 24-cell shell Laplacians on each coset zero-extend into eigenspaces of the global 600-cell Laplacian. It proves that the $\lambda=12$ local eigenspaces do so exactly, and that no other local eigenvalue admits a non-trivial lift.

That spectral result is independent and is not reproved here; the two artifacts can be read in either order, but the finite-geometric decomposition certified in this note is logically prior.

8 Reproducibility

One-command reproduction.

```
python closure_transform_engine/examples/run_wo007_schlafl_i_decomposition.py
pytest closure_transform_engine/tests/test_wo007_schlafl_i_decomposition.py
```

The first command (under a minute) builds V_{600} from exact $\mathbb{Q}(\sqrt{5})$ icosian quaternions, runs every certificate, and writes the frozen output artifacts below. The second re-asserts the certificate-level claims as a test.

Environment. Pinned in `requirements.txt` and `pyproject.toml`: Python ≥ 3.8 (developed on 3.8.10), NumPy ≥ 1.20 (1.24.4), SymPy ≥ 1.12 (1.13.3). All formal certificates use exact arithmetic via `vfd_v600` ($\mathbb{Q}(\sqrt{5})$ -rational pairs) and integer set operations on vertex indices; NumPy and SymPy enter only auxiliary computations and never a formal claim. The reproducibility log `wo007_reproducibility_log.txt` records UTC timestamp, elapsed time, Python and dependency versions, platform, commit hash, and per-certificate pass/fail states.

Dependency chain. The claims of this note do not require any external repository. A reader can verify every theorem by running the two commands above. The vendored `vfd_v600` package supplies exact $\mathbb{Q}(\sqrt{5})$ icosian arithmetic; it is the only dependency beyond the standard scientific Python stack.

What the runner verifies:

- Theorem 1 (exact $V_{600} = 2I$): 120 distinct unit-norm vertices, closure under multiplication, identity and inverses present.
- Theorem 2 ($V_{24} = 2T$): exactly 24 σ -fixed vertices, subgroup property, index 5 in $2I$.
- Theorem 3 (Schläfli decomposition): five right cosets, each of size 24, pairwise disjoint, union equals V_{600} , $C_0 = 2T$.
- Theorem 4 (each coset is a 24-cell): shortest intra-coset d^2 is exactly 1, 8-regular, 96 edges, and no intra-coset V_{600} -edge.
- Theorem 6 (A_5 coset action): 60 distinct permutations, kernel exactly $\{\pm 1\}$, image all even with class sizes $(1, 15, 20, 12, 12)$, hence A_5 .

Outputs (under the docs outputs directory):

- `wo007_summary.json` — full structured certificate result.
- `wo007_cosets.json` — the five coset vertex-index lists.
- `wo007_coset_action_table.csv` — for each $h \in 2I$, the induced coset permutation, its cycle structure, identity flag, and parity.
- `wo007_distance_shells.csv` — the eight non-zero V_{600} -distance shells and their pair counts.
- `wo007_reproducibility_log.txt` — run UTC timestamp, elapsed time, environment, commit hash, and per-certificate pass/fail states.

References

- [Coxeter 1973] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973.
- [Conway–Sloane 1999] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer-Verlag, 1999.
- [Conway et al. 1985] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.
- [Smart 2026] L. Smart, *The 24–600 Spectral Bridge: A selective $\lambda=12$ eigenspace embedding from five 24-cell cosets into the 600-cell*. Public reproducible note, Institute of Vibrational Field Dynamics, 2026. Repository: <https://github.com/vfd-org/the-24-600-spectral-bridge>.